



Smooth operator: Rigorous training, starting at an early age, has enabled top sliders like Novak Djokovic to move confidently and effortlessly.

move towards the ball in this position. Most shots require a hitting position. This position accomplishes two things: first, it allows the player to hit powerful shots or to shorten their stroke to deal with fast balls. Second, it immediately completes much of the racket's path, leaving the player with more time to deal with the oncoming shot. Players can pause in the hitting position or simply slow down before completing the stroke. If a player is late getting into — or passing through — the hitting position, their only option may be to speed up their stroke, which makes errors more likely.

Improve your footwork to improve your timing

A properly timed and correctly executed split step is essential for

excellent ball timing, stresses BJ Miller, a biomechanics expert on the split step in tennis: "The split step is the first move in executing a shot," said Miller. "Often characterised as a single, instantaneous move, it actually has parts: the take-off, the flight, the touchdown, and the knee flex after touchdown. "The split step has four purposes: 1) It allows players to come into balance. 2) It results in a flexing of the knees to produce a forceful first move to the ball. 3) It results in storage of elastic energy in the player's legs to make the first move to the ball even more forceful. 4) It allows a player to make their first move towards the oncoming ball as soon as the player detects the ball's direction, roughly 0.2 seconds.

"The height of the split step should be approximately three inches," continued Miller. "Much higher split steps take too much time. The knees should be flexed at least 70° — the angle of the front of the lower leg to the front of the thigh

— just before landing to get the maximum push-off for the first move to the oncoming ball. Upon landing, the knees should flex somewhat more so the total knee flex is at least 70°. The landing should occur at the instant of the opponent's ball contact. The direction of the oncoming ball can be detected in roughly 0.2 seconds which is, fortuitously, roughly the duration of the after-landing knee flex. Therefore, the first move to the ball can be initiated roughly 0.2 seconds after the opponent hits it."

Arrive with a well-timed, controlled slide: Rigorous training, starting at an early age, has enabled today's top sliders to move confidently and effortlessly. From Bjorn Borg and Guillermo Vilas in the 1970s to Nadal and Alcaraz this century, clay-court champions have mastered the art of sliding. As these maestros slide and glide towards the oncoming ball, their movement then links with and helps initiate their body's kinetic chain, which unleashes